



Cloud Computing

Assignment-Week 4

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 10

Total mark: 10 X 1 = 10

QUESTION 1:

Which Azure SQL Database service tier separates compute and storage so that storage can automatically grow up to 100 TB without requiring application downtime?

- A. Basic
- B. Business Critical
- C. Hyperscale
- D. Standard

Correct Answer: C

Detailed Solution: The Hyperscale tier decouples compute and storage, allowing storage to automatically scale up to 100 TB. It is intended for very large databases and supports rapid scaling and fast backup/restore operations.

QUESTION 2:

One company deploys two Azure Virtual Machines in different Availability Zones within the same Azure region. One data center experiences a complete power outage. Which statement is correct?

- A. Both VMs will become unavailable because they are in the same region.
- B. The second VM can continue running because Availability Zones are isolated from each other.
- C. Azure automatically moves the failed VM to another region.
- D. Both VMs share the same cooling and networking infrastructure.

Correct Answer: B

Detailed Solution: Availability Zones are physically separate data centers within the same Azure region, each with independent power, cooling, and networking. If one zone fails, resources in another zone continue to operate.

QUESTION 3:

Which App Service pricing tier supports the highest number of deployment slots?

- A. Basic
- B. Standard



- C. Premium (P1v3 and higher)
- D. Free

Correct Answer: C

Detailed Solution: The Premium (P1v3 and higher) App Service plan supports up to 20 deployment slots, making it suitable for advanced DevOps and CI/CD scenarios.

QUESTION 4:

An administrator notices that users can still access an Azure App Service website using http://. Which App Service configuration should be enabled to prevent unencrypted access?

- A. Managed Identity
- B. Always On
- C. HTTPS Only
- D. Deployment Slots

Correct Answer: C

Detailed Solution: Enabling HTTPS Only forces all incoming HTTP requests to be redirected to HTTPS, preventing unencrypted communication.

QUESTION 5:

Which Google Cloud database service is designed specifically for structured relational data with global consistency and horizontal scalability?

- A. Firestore
- B. Cloud SQL
- C. Cloud Spanner
- D. Memorystore

Correct Answer: C

Detailed Solution: Cloud Spanner is a globally distributed relational database that combines SQL support, horizontal scalability, and strong consistency. Firestore is NoSQL, Cloud SQL scales primarily vertically, and Memorystore is an in-memory cache.

QUESTION 6:

A virtual machine requires an additional 100 GB persistent disk that should remain intact even after the VM is terminated. Which OpenStack service manages this storage?



- A. Nova
- B. Swift
- C. Cinder
- D. Keystone

Correct Answer: C.

Detailed Solution: Cinder provides persistent block storage volumes independent of the VM lifecycle. If the VM is deleted, the Cinder volume remains unless explicitly removed.

QUESTION 7:

Which statement correctly distinguishes GKE from Cloud Run?

- A. GKE only supports virtual machines.
- B. Cloud Run requires users to manage Kubernetes clusters.
- C. GKE manages Kubernetes clusters, while Cloud Run executes containers without requiring cluster management.
- D. Both services require manual Kubernetes control plane management.

Correct Answer: C

Detailed Solution: GKE is a managed Kubernetes service, whereas Cloud Run is a serverless container platform where Google manages the underlying infrastructure and scaling.

QUESTION 8:

A company collects millions of IoT sensor events every minute. The data must be ingested in real time, stored in a scalable analytical warehouse, and visualized through interactive dashboards.

Which GCP architecture best satisfies these requirements?

- A. Cloud SQL + Cloud Functions + Looker
- B. Pub/Sub + BigQuery + Looker Studio
- C. Firestore + Compute Engine + Bigtable
- D. Cloud Storage + Memorystore + App Engine

Correct Answer: B



Detailed Solution: Pub/Sub provides scalable real-time event ingestion. BigQuery stores petabyte-scale analytical data and supports SQL queries. Looker Studio connects directly to BigQuery for dashboards and reports. Firestore is operational NoSQL storage, while Cloud SQL is not suitable for high-volume analytics.

QUESTION 9:

A DevOps engineer wants to route all production traffic to version v3 of an App Engine service. Which command is correct?

- A. `gcloud app services set-traffic default --splits=v3=1`
- B. `gcloud app deploy --traffic=v3`
- C. `gcloud app versions promote v3`
- D. `gcloud app services update v3`

Correct Answer: A

Detailed Solution: Setting the split to v3=1 directs 100% of traffic to version v3.

QUESTION 10:

Which OpenStack service is primarily responsible for routing tenant traffic between internal and external networks?

- A. Nova
- B. Neutron
- C. Keystone
- D. Swift

Correct Answer: B

Detailed Solution: Neutron provides networking services, including virtual networks, routers, DHCP, security groups, and floating IPs.
